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Interactive effects of green-space exposure and air pollution on cardiovascular disease risk: a study based on the CHARLS and CLHLS databases

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Abstract

Background Cardiovascular diseases are common in aging populations and are closely associated with multiple factors. However, the cumulative effects of green-space exposure and air pollutants on cardiovascular disease have not been adequately investigated.

Methods This study used CHARLS 2015 data and CLHLS 2014 data. Green-space exposure was quantified using the normalized difference vegetation index (NDVI), while exposures to PM_{2.5}, NO₂ and SO₂ were assessed using the China High-resolution Air Pollutant dataset (CHAP). A composite cardiometabolic risk score was derived from five metabolic syndrome (MetS) diagnostic components: abdominal obesity, hypertension, hyperglycemia, hypertriglyceridemia, and low high-density lipoprotein. Generalized linear models (GLMs) were applied to evaluate the effects of green-space exposure and air pollutants on cardiovascular disease risk; single-, two-, and three-pollutant models were fitted, and interaction and sensitivity analyses were conducted to assess the robustness of the findings.

Results Green-space exposure was associated with a lower risk of cardiovascular disease (OR = 0.867, 95% CI: 0.796–0.944, $P = 0.001$), while exposures to PM_{2.5}, NO₂, and SO₂ were all linked to an increased risk. Among these pollutants, NO₂ showed the strongest association, with each interquartile range (IQR) increase corresponding to a 25.9% higher risk of cardiovascular disease (OR = 1.259, 95% CI: 1.152–1.377, $P < 0.001$). Interaction analysis revealed that higher green-space exposure had a more pronounced protective effect under low levels of air pollution. However, under high pollution levels, greater green-space exposure appeared to amplify the adverse effects of pollutants on cardiovascular disease risk. In the multi-pollutant model, after further adjusting for three pollutants, the association between PM_{2.5} (OR = 1.179, 95% CI: 1.017–1.366, $P = 0.028$) and NO₂ (OR = 1.165, 95% CI: 1.012–1.341, $P = 0.034$) and cardiovascular disease risk remained significant.

Conclusions Evidence indicates that green-space exposure confers protection for cardiovascular health, but this effect is significantly modulated by ambient air pollution.

Keywords Cardiovascular disease, Green space exposure, Air pollutants, Adults and elderly diseased persons

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Background

Cardiovascular diseases (CVDs), encompassing conditions such as coronary heart disease, cerebrovascular disease, and peripheral artery disease, represent a group of chronic disorders affecting the heart and vascular system. They are characterized by high morbidity, disability, and mortality rates worldwide [1]. As the leading cause of death globally, CVDs not only threaten individual health but also impose a substantial burden on public health systems [2]. According to the World Health Organization (WHO) Global Cardiovascular Disease Report 2024, CVDs are responsible for approximately 17.9 million deaths each year, accounting for 31.8%–32% of all global deaths, with about one-quarter of these deaths attributable to environmental factors [3].

The impact of CVDs extends widely, contributing to both acute events—such as myocardial infarction and heart failure—and long-term health complications, which are particularly severe among older adults. At the cognitive level, CVDs can lead to cognitive decline, memory impairment, and deficits in executive function, thereby increasing the risk of dementia [4]. Psychologically, patients with cardiovascular disease often experience emotional disorders such as anxiety and depression, forming a vicious cycle through brain–heart interactions [5]. From a socioeconomic perspective, the long-term treatment and care required for CVDs consume significant medical resources, imposing ongoing financial pressure on patients and their families. Additionally, CVDs can limit participation in social activities, leading to social isolation and worsening health outcomes [6]. The resulting loss of labor capacity and decline in productivity further pose serious challenges to national economic development and social stability [7].

Among the numerous factors influencing the development of CVDs, environmental exposures play a critical role. Green-space exposure and air pollutants represent two interrelated yet mechanistically distinct environmental factors. Green-space exposure refers to the extent of contact with natural vegetation, typically quantified using indicators such as the Normalized Difference Vegetation Index (NDVI) [8]. Exposure to green environments may promote cardiovascular health through multiple pathways, including providing spaces for physical activity, reducing psychological stress, improving air quality, and fostering social interaction [9].

Air pollutants, on the other hand, refer to harmful gases and particulate matter present in the environment—mainly PM_{2.5}, NO₂, and SO₂—which can impair the cardiovascular system through mechanisms such as oxidative stress, inflammation, and autonomic dysfunction [10].

Although previous studies have examined the effects of green-space exposure and air pollution on cardiovascular

health, the potential interactions between these two environmental factors remain insufficiently explored. In the context of rapid urbanization, urban greening and air pollution control are often considered separately, lacking integrated approaches. Exploring the interactive effects between green-space exposure and air pollutants helps to clarify the complex mechanisms underlying environmental influences on cardiovascular health and provides scientific evidence to support more precise and effective environmental intervention strategies.

Green space may modify the health effects of air pollution through both protective and adverse pathways. Exposure to green space can attenuate pollution-related cardiovascular risk by reducing stress, promoting physical activity, and improving cardiometabolic resilience, thereby lowering susceptibility to pollution-induced inflammation and oxidative stress. Vegetation may also remove some airborne particles through deposition. Conversely, green space may amplify pollution-related health risks in certain contexts: dense vegetation near emission sources can limit air dispersion and increase local pollutant concentrations, and biogenic volatile organic compounds emitted by some plants may contribute to the formation of secondary pollutants. Green spaces may also encourage longer outdoor exposure, increasing inhaled pollutant doses in highly polluted environments. We therefore hypothesize that green space attenuates the association between air pollution and cardiovascular disease risk in low-pollution settings, but may amplify pollution-related health risks under high-pollution conditions.

Methods

Study population

Data were drawn from two large national cohorts: the China Health and Retirement Longitudinal Study (CHARLS) and the Chinese Longitudinal Healthy Longevity Survey (CLHLS). CHARLS, initiated in 2011, covers approximately 17,000 individuals from ~ 10,000 households across 150 county-level and 450 village-level units in China, and provides rich data on socioeconomic status, health conditions, and behavioral factors. CLHLS, launched in 1998, is one of the largest and longest-running nationwide databases on older adults' health in China [11–12]. Combining these two databases offers sufficient sample size to support analyses of complex interaction effects and, by leveraging multidimensional covariates, helps reduce confounding bias when assessing the joint impact of green-space exposure and air pollutants on cardiovascular outcomes.

The analysis used CHARLS data from 2015 and CLHLS data from 2014. Participants with missing information on province, age, sex, depression status, body mass index (BMI), physical activity or cardiovascular variables were

excluded, as were individuals younger than 45 years in CHARLS and younger than 45 years in CLHLS. After these exclusions, 11,505 participants from CHARLS and 1,398 participants from CLHLS were retained, yielding a combined analytic sample of 12,903 individuals.

Exposure

Green-space exposure was assessed using the Normalized Difference Vegetation Index (NDVI), obtained from the global MODIS database (MOD13Q1, Version 5). Annual average NDVI values (multiplied by 10,000) were used to estimate individual exposure levels, minimizing the influence of seasonal variation. Indeed, the NDVI value is between 0 and 1, but in order to avoid accuracy loss caused by decimals and facilitate data calculation, we multiplied the NDVI value by 10,000 and rounded it to obtain the actual NDVI value used. At the same time, existing studies have confirmed that quantitative NDVI is consistent with on the ground green ness measurements [13]. Air pollutants (PM_{2.5}, NO₂, SO₂) were obtained from the China High-resolution Air Pollutant (CHAP) dataset, which integrates ground observations, satellite products, reanalysis and model simulations with AI to account for spatio-temporal heterogeneity. Annual average concentrations for each participant were calculated using the same spatial estimation method applied to green-space data. The spatial resolution for each pollutant was 1*1 square kilometers. Cross-validated coefficients of determination (CV-R²) were 0.92 for PM_{2.5}, 0.93 for NO₂, and 0.84 for SO₂. For CHARLS participants, environmental exposure was assigned using the 2015 annual average value, corresponding to the survey waves assessing cardiovascular risk components. For CLHLS participants, exposure levels are based on annual averages consistent with the 2014 CLHLS survey year. Due to the fact that the resident information in the CHARLS and CLHLS databases used in this study is only available at the provincial level, all exposure indicators mentioned above are assigned to all participants by province (in CLHLS, respondents directly report their province of residence in the questionnaire; In CHARLS, each participant is assigned a unique identification code, where the first two digits correspond to the official provincial administrative code).

Cardiovascular composite score

The cardiometabolic composite score was developed based on the diagnostic criteria for metabolic syndrome (MetS), which is a validated predictor of future cardiovascular disease risk [14]. The criteria integrate five key metabolic abnormalities: abdominal obesity, hypertension, hyperglycemia, hypertriglyceridemia, and low high-density lipoprotein cholesterol (HDL-C). The specific definitions are as follows:

- (1) Abdominal obesity: waist circumference ≥ 90 cm for men and ≥ 80 cm for women.
- (2) Hypertension: current use of antihypertensive medication, or an average systolic blood pressure ≥ 130 mmHg or diastolic blood pressure ≥ 85 mmHg measured across three readings.
- (3) Hyperglycemia: current use of glucose-lowering medication or insulin, fasting blood glucose ≥ 110 mg/dL (6.1 mmol/L), or glycated hemoglobin (HbA1c) $\geq 6.5\%$.
- (4) Hypertriglyceridemia: serum triglycerides ≥ 150 mg/dL (1.7 mmol/L).
- (5) Low HDL-C: <40 mg/dL (1.0 mmol/L) for men and <50 mg/dL (1.3 mmol/L) for women.

Participants meeting three or more of these criteria were classified as high-risk; those meeting fewer than three were considered low-risk.

Covariates

A series of potential confounders were included in the analysis and categorized as follows. Demographic characteristics comprised sex, age, marital status (married or unmarried), and residential environment (urban or rural). Health-related behaviors included smoking status (never, former, current), alcohol consumption (never, former, current), physical activity (high activity level, low activity level), where engaging in vigorous activity at least once a week is defined as high activity level and sleep duration: we consider a sleep duration of no more than 4 h as severe deficiency, 4–7 h (excluding 7 h) as insufficient, 7–9 h (excluding 9 h) as normal, 9–12 h as excessive, and exceeding 12 h (including 12 h) as extreme excess. Health status variables included self-rated health (poor, fair, good, very good, or excellent), depression status (none, mild, or severe), and body mass index (BMI), calculated as weight (kg) divided by height squared (m²) and classified into underweight, normal weight, and overweight categories.

Statistical analysis

Participants were classified into low- and high-cardiovascular-risk groups and described using descriptive statistics. Categorical variables are presented as counts (*n*) and percentages (%), and represent continuous variables as mean (95% CI). Differences between the high-risk and low-risk groups were assessed using the Student's *t*-test for continuous variables and the chi-square test for categorical variables. *P* values were used to evaluate the association between each covariate and cardiovascular risk level.

Generalized linear models (GLMs) were fitted to examine the associations of each air pollutant and green-space exposure with cardiovascular risk. The linking function

employs logistic regression; odds ratios (ORs) and 95% confidence intervals (95% CIs) were reported. For exposure scaling, the interquartile range (IQR) was calculated for each pollutant and for NDVI; exposures were also categorized into quartiles (Q1–Q4) to assess the effect of an IQR (or quartile) increase on cardiovascular risk.

Table 1 Baseline data

		Low Risk(10240)	High Risk(2663)	P- value
Age (95%CI)		63.1 (58.6–67.6)	61.3 (57.3–65.2)	< 0.001
Gender (%)	Male	5347 (52.2)	841 (31.6)	< 0.001
	Female	4893 (47.8)	1822 (68.4)	
Marital Status (%)	Unmarried	1777 (17.4)	380 (14.3)	< 0.001
	Married	8463 (82.6)	2283 (85.7)	
Living Environ- ment (%)	Urban	1986 (19.4)	705 (26.5)	< 0.001
	Rural	8254 (80.6)	1958 (73.5)	
Smoking History (%)	Never Smoked	5887 (57.5)	1915 (71.9)	< 0.001
	Former Smoker	1726 (16.9)	374 (14.0)	
	Current Smoker	2627 (25.7)	374 (14.0)	
Alcohol Con- sumption (%)	Never Drank Alcohol	6734 (65.8)	2072 (77.8)	< 0.001
	Former Drinker	253 (2.5)	54 (2.0)	
	Current Drinker	3253 (31.8)	537 (20.2)	
Self-Rated Health (%)	Poor	932 (9.1)	326 (12.2)	< 0.001
	Fair	3392 (33.1)	930 (34.9)	
	Good	3508 (34.3)	847 (31.8)	
	Very Good	1703 (16.6)	378 (14.2)	
	Excellent	705 (6.9)	182 (6.8)	
Sleep Duration (%)	Severely	756 (7.4)	206 (7.7)	0.881
	Insufficient	4290 (41.9)	1092 (41.0)	
	Normal	4047 (39.5)	1073 (40.3)	
	Excessive	1062 (10.4)	270 (10.1)	
	Abnormally	85 (0.8)	22 (0.8)	
Depression(%)	No Depression	6851 (66.9)	1813 (68.1)	0.476
	Mild Depression	2872 (28.0)	725 (27.2)	
	Severe Depression	517 (5.0)	125 (4.7)	
BMI (%)	Underweight	800 (7.8)	16 (0.6)	< 0.001
	Normal	5646 (55.1)	553 (20.8)	
	Overweight	3794 (37.1)	2094 (78.6)	
Physical Activity (%)	Low activity level	7454 (72.8)	2122 (79.7)	< 0.001
	High activity level	2786 (27.2)	541 (20.3)	

Three sequential models were specified: Model 1 was unadjusted; Model 2 adjusted for demographic factors (age, sex, marital status, and residential setting); and Model 3 additionally adjusted for health behaviors (smoking, alcohol consumption, physical activity and sleep duration) and health status (self-rated health, depression, and BMI). Analyses were performed separately in the CHARLS and CLHLS datasets to assess the consistency of results across cohorts.

To investigate joint effects, NDVI and air-pollutant exposures were standardized and interaction analyses were conducted. Multipollutant models, including two-pollutant and three-pollutant specifications, were used to evaluate combined pollutant effects; these models were adjusted using the full set of covariates in Model 3.

Several sensitivity analyses were conducted to test robustness:

- (1) Stratification by residential region (eastern, central, western provinces) to address potential regional heterogeneity;
- (2) Exclusion of individuals who had moved residence ($n = 780$) to remove bias related to migration;
- (3) To eliminate the potential impact of ambient temperature on experimental results, the annual average temperature of each subject's province was calculated, using the same method as for NDVI and air pollutants.
- (4) Use of one-year lagged annual average pollutant concentrations to examine potential lagged effects of air pollution.

All statistical analyses were performed in R version 4.4.1, and two-sided P values < 0.05 were considered statistically significant.

Results

Descriptive statistics

A total of 12,903 participants were included in the analysis and were categorized into low- and high-cardiovascular-risk groups (Table 1). Continuous variables are reported as means (95% CI) and categorical variables as counts (n) and percentages (%). The low-risk group comprised 10,240 individuals, while the high-risk group included 2,663 individuals. The mean age was 63.1 years in the low-risk group and 61.3 years in the high-risk group. Compared with the low-risk group, individuals in the high-risk group were more likely to be female, married, urban residents, never smokers, never drinkers, to report poor self-rated health, to be overweight, and low activity level (all $P < 0.001$). Although participants with severely insufficient sleep and those without depression

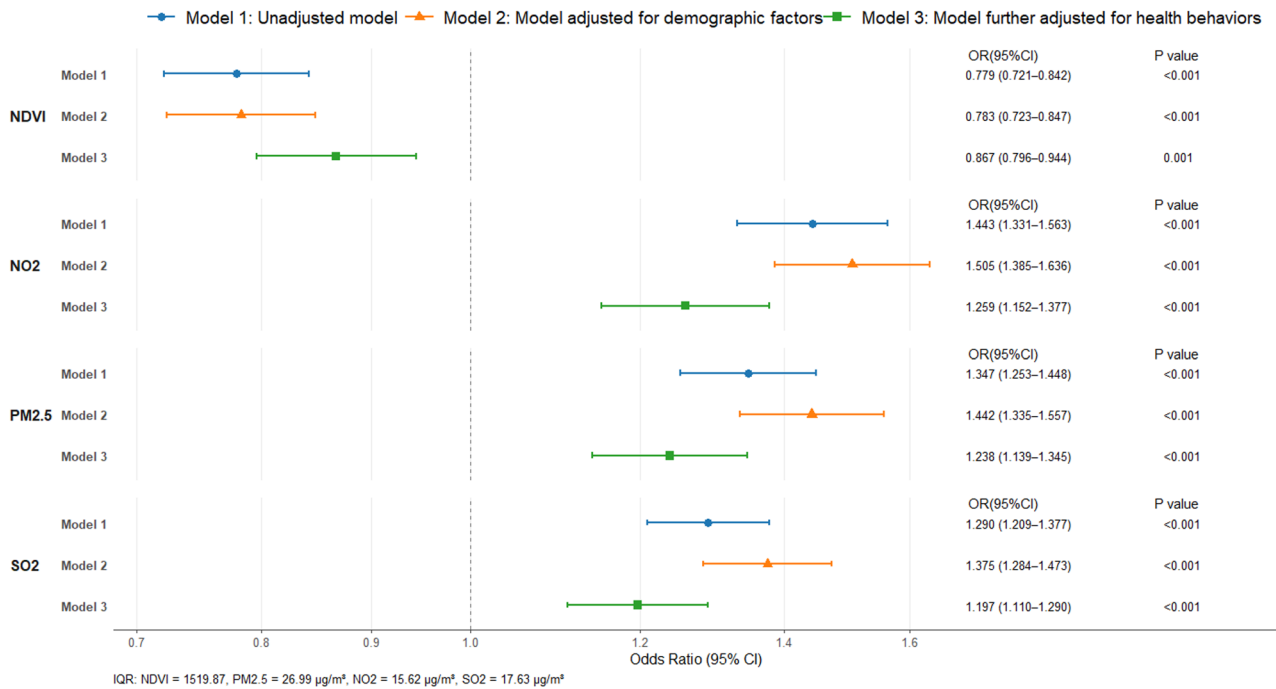


Fig. 1 Green-space exposure and air pollutants in relation to cardiovascular disease risk

Table 2 Group analysis by database

Data Base	Head	Environment Variable	OR (95%CI)	P
CLHLS	1398	Green space exposure	0.690 (0.398–1.194)	0.184
		PM2.5	2.080 (1.544–2.865)	<0.001
		NO ₂	1.990 (1.454–2.773)	<0.001
		SO ₂	1.567 (1.282–1.926)	<0.001
CHARLS	11,505	Green space exposure	0.864 (0.792–0.942)	<0.001
		PM2.5	1.210 (1.104–1.325)	<0.001
		NO ₂	1.230 (1.118–1.353)	<0.001
		SO ₂	1.175 (1.079–1.279)	<0.001

showed higher proportions in the high-risk group, these differences did not reach statistical significance ($P>0.05$).

The association between air pollution, green space exposure, and cardiovascular disease risk

The associations between each air pollutant, green-space exposure, and cardiovascular disease risk are presented in Fig. 1. The interquartile range (IQR) increments for green-space exposure, PM_{2.5}, NO₂, and SO₂ were 1,519.87, 26.99 µg/m³, 15.62 µg/m³, and 17.63 µg/m³, respectively. Across all models—including the unadjusted Model 1 and the adjusted Models 2 and 3—green-space exposure was associated with a reduced risk of cardiovascular disease, whereas PM_{2.5}, NO₂, and SO₂ were associated with increased risk. Among the pollutants, NO₂ showed the strongest association, with each IQR increase

corresponding to a 27.2% higher risk of cardiovascular disease (OR = 1.259, 95% CI: 1.152–1.377, $P<0.001$).

Analyses were further stratified by data source into CHARLS and CLHLS cohorts (Table 2). The associations between air pollutants and increased cardiovascular risk were consistent across both databases. However, the protective effect of green-space exposure on cardiovascular risk was not statistically significant in the CLHLS cohort ($P=0.184$), likely due to the smaller sample size in that database.

The combined effect of green space exposure and air pollutants on cardiovascular disease risk

Interaction analyses were conducted to further assess the combined effects of green-space exposure and air pollutants on cardiovascular disease risk, with results presented in Fig. 2. In Fig. 2, the low level is the mean minus standard deviation, the mean level is the mean, and the high level is the mean plus standard deviation. The analyses indicated a consistent pattern: under low levels of air pollution, higher green-space exposure provided stronger protective effects, mitigating the impact of pollutants on cardiovascular risk. In contrast, at higher pollution levels, greater green-space exposure appeared to amplify the adverse effects of air pollutants on cardiovascular risk. The specific effects are shown in Table 3.

Multi-pollutant analysis

To evaluate the combined impact of multiple pollutants, single-, two-, and three-pollutant models were analyzed

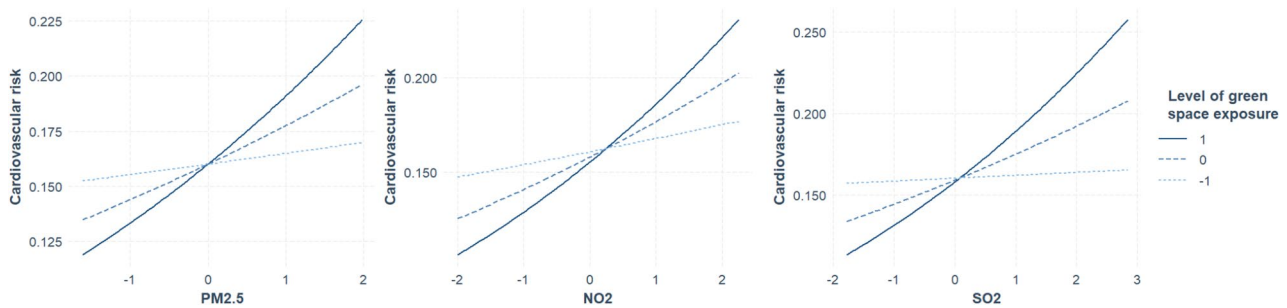


Fig. 2 Interaction effects between green-space exposure and air pollutants on cardiovascular disease risk

Table 3 Interaction effect

Green Space Value		OR(95%CI)	P
NDVI * PM2.5		1.093 (1.033–1.157)	0.002
PM2.5	Low Level	1.037(0.964–1.115)	
	Medium Level	1.133 (1.072–1.198)	
	High Level	1.239 (1.137–1.350)	
NDVI * SO2		1.107 (1.041–1.177)	0.001
SO2	Low Level	1.013 (0.943–1.089)	
	Medium Level	1.121 (1.066–1.180)	
	High Level	1.241 (1.138–1.354)	
NDVI * NO2		1.087 (1.035–1.141)	<0.001
NO2	Low Level	1.053 (0.991–1.118)	
	Medium Level	1.143 (1.085–1.206)	
	High Level	1.243 (1.145–1.349)	

Table 4 Multi-pollutant analysis

	OR(95%CI)	P
PM2.5	1.259 (1.152–1.377)	<0.001
PM2.5 + NO ₂	1.133 (0.908–1.413)	0.267905
PM2.5 + SO ₂	1.108 (0.883–1.391)	0.3750568
PM2.5 + NO ₂ + SO ₂	1.179 (1.017–1.366)	0.02845065
NO ₂	1.238 (1.139–1.345)	<0.001
NO ₂ + PM2.5	1.114 (0.907–1.367)	0.3041419
NO ₂ + SO ₂	1.080 (0.869–1.343)	0.4863768
NO ₂ + PM2.5 + SO ₂	1.165 (1.012–1.341)	0.03422781
SO ₂	1.197 (1.110–1.290)	<0.001
SO ₂ + PM2.5	1.072 (0.948–1.213)	0.2663134
SO ₂ + NO ₂	1.071 (0.942–1.216)	0.292513
SO ₂ + PM2.5 + NO ₂	1.056 (0.926–1.204)	0.4151758

(Table 4). In single-pollutant models, PM_{2.5}, NO₂, and SO₂ were each associated with increased cardiovascular risk ($P < 0.001$). In three-pollutant models, the associations of PM_{2.5} (OR = 1.179, 95% CI: 1.017–1.366, $P = 0.028$) and NO₂ (OR = 1.165, 95% CI: 1.012–1.341, $P = 0.034$) with cardiovascular risk remained significant. Associations in other two-pollutant combinations and in three-pollutant models were no longer statistically significant.

Sensitivity analysis

The associations between air pollutants and cardiovascular disease risk remained consistent and statistically

significant ($P < 0.001$) across all sensitivity analyses, indicating the robustness of these findings (Fig. 3). After excluding participants with changes in residence and further adjusting the residential area and ambient temperature in Model 3, the effects of pollutants on cardiovascular risk were slightly reduced. In addition, pollutant exposures were replaced with one-year lagged annual average concentrations to evaluate potential lagged effects.

Discussion

Theoretical validation of the independent effects of green space exposure and air pollutants

Evidence indicates a negative association between green-space exposure and cardiovascular disease risk [15]. Long-term follow-up of large populations has shown that residents living in areas with higher levels of greenery experience lower incidence of cardiovascular events, with a 14.3% reduction in the likelihood of hospitalization due to cardiovascular disease or stroke. Even after considering potential confounding factors such as nighttime lighting, air pollution, and noise in the model, this protective effect and correlation still exist. Furthermore, quantification of the relationship between green-space exposure and cardiovascular health demonstrates the multi-level protective mechanisms of green environments on the cardiovascular system [16], providing more precise reference thresholds for evidence-based intervention strategies.

Regarding air pollutants, studies have consistently shown that PM_{2.5}, NO₂, and SO₂ are positively associated with increased cardiovascular disease risk, confirming a significant link between environmental pollutant exposure and elevated incidence of hypertension [17]. Notably, the effect size of NO₂ observed in this study was higher than that of PM_{2.5} and SO₂, which may be attributed to its higher biological activity and greater ability to penetrate alveoli into the bloodstream [18]. In addition, NO₂ primarily originates from vehicle exhaust and may act synergistically with other traffic-related pollutants, such as ultrafine particles and polycyclic aromatic

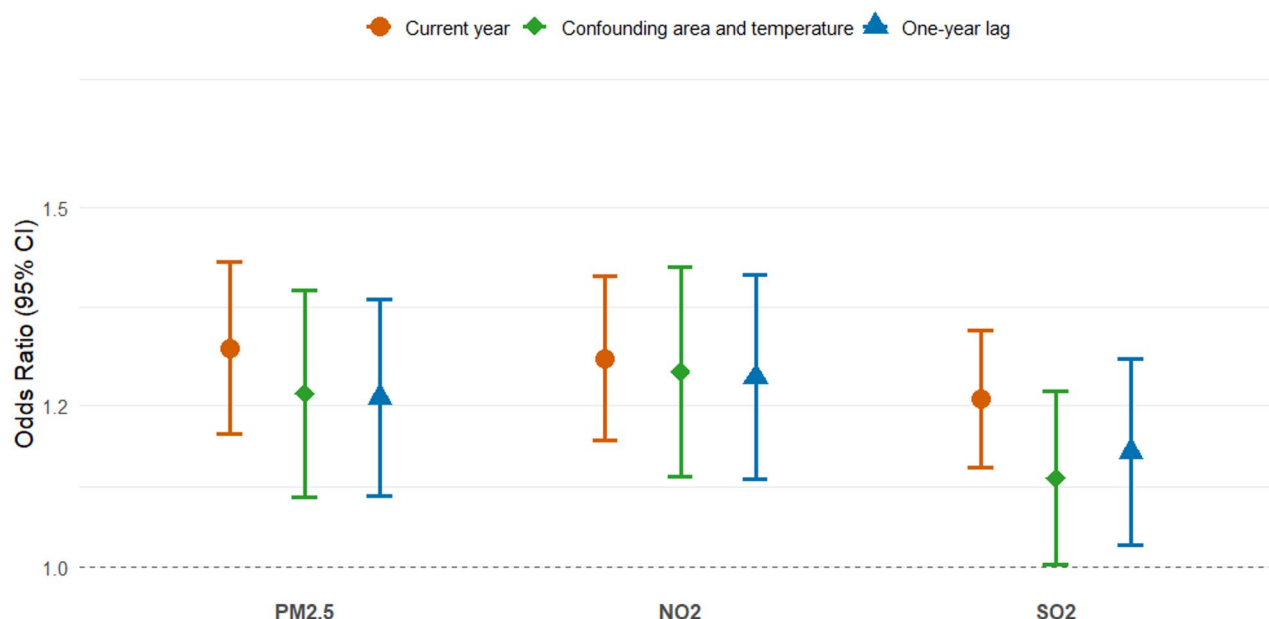


Fig. 3 Comparison of immediate and lagged effects of air pollutants on cardiovascular disease risk

hydrocarbons, thereby exacerbating cardiovascular damage [19].

Discovery of the interaction effects between green space and pollution

The findings reveal a nonlinear interaction between green-space exposure and air pollutants: in low-pollution environments, green-space exposure provides clear protective effects, whereas in high-pollution environments, higher green-space exposure is associated with elevated cardiovascular risk.

This observation challenges the conventional view that vegetation reduces pollutant concentrations primarily through deposition [20]. Several potential explanations for this phenomenon are proposed:

- (1) High vegetation coverage may increase air turbulence and surface deposition, leading to higher localized pollutant accumulation within green spaces [21].
- (2) Certain plants in highly polluted environments may emit volatile organic compounds (VOCs), which can react with air pollutants to form secondary pollutants (e.g., ozone, O_3) [22], thereby increasing health risks.
- (3) Green spaces in heavily polluted areas may attract more residents for outdoor activities, prolonging exposure time and increasing pollutant dose, which could offset the inherent health benefits of vegetation.

Of particular note, the interaction between NO_2 and green-space exposure was the most pronounced. This may relate to the unique physicochemical properties of NO_2 . Its high water solubility allows it to be readily absorbed by plant leaves, potentially forming nitrate particles on vegetation surfaces [23]. When wind passes over these green areas, pollutants adhering to leaves may become resuspended and inhaled by humans. This mechanism provides micro-level theoretical support for understanding why green-space exposure may exacerbate cardiovascular risk in highly polluted environments.

Expansion of the multi-pollutant model

Analyses using single-, two-, and three-pollutant models provided insights into the combined effects of multiple pollutants on cardiovascular health. In single-pollutant models, $PM_{2.5}$, NO_2 , and SO_2 were each significantly associated with increased cardiovascular disease risk. However, in multipollutant models, only $PM_{2.5}$ and NO_2 remained significantly associated, suggesting that these two pollutants may affect the cardiovascular system through independent and complementary pathological mechanisms.

$PM_{2.5}$ primarily induces systemic inflammation and oxidative stress via pulmonary deposition, impairing vascular endothelial function and blood rheology [24]. NO_2 , as a gaseous pollutant, may more directly influence autonomic regulation and cardiac electrophysiological properties [25]. The synergistic effects of these two pollutants could lead to more severe cardiovascular damage, explaining why their associations remain significant even after adjusting for other pollutants. In contrast, the effect

of SO₂ may be partly masked by PM_{2.5} and NO₂, as SO₂ readily converts to sulfate particles in the atmosphere, and its health impact may already be captured within the PM_{2.5} effect.

Exposure to multiple pollutants may be associated with greater cardiovascular risk than exposure to single pollutants. Coordinated reduction strategies targeting both PM_{2.5} and NO₂ are needed. Considering that NO₂ primarily originates from motor vehicle emissions, while PM_{2.5} has more diverse sources [26], interventions in urban traffic management and industrial emission control should be prioritized.

Theoretical paradigm shift driven by environmental interaction effects

By revealing a non-linear interaction between green-space exposure and air pollutants on cardiovascular disease, the research offers a new perspective for environmental-health theory. Traditional models in environmental health typically employ single-exposure, single-outcome linear frameworks and treat green space and air pollution as independent factors. [27, 28] Our findings demonstrate a significant interaction between the two exposures and identify a potential risk-amplifying effect of green space under high pollution conditions. This result challenges the simplistic assumption that more green space is always better and supports a shift from linear to complex-systems models in environmental-health theory.

Traditional environmental risk-assessment models (for example, the EPA's air-quality health indices) focus primarily on pollutant concentration–response relationships and do not emphasize how protective environmental features such as green space may modify those relationships [29]. The interaction detected here suggests a pathway toward a more precise framework for environmental-health risk assessment and aligns with the view that people are chronically exposed to complex mixtures of pollutants and chemicals [30]. Importantly, our large-sample empirical analysis identifies concrete threshold values for these interactions, providing actionable detail beyond conceptual recognition of mixture effects.

The non-linear correlation reported here suggests that future research should investigate how local pollution background affects the relationship between green spaces and cardiovascular health outcomes, in order to further optimize health outcomes. For instance, indiscriminately increasing green-space area in heavily polluted zones could be counterproductive, whereas in low-pollution areas expanding green space is likely to maximize health benefits. Planning and intervention strategies therefore need to be tailored to the ambient pollution background rather than applied uniformly.

From an applied planning perspective, urban green-space design has long prioritized aesthetics and recreation; the present results indicate that health benefits of green space are strongly modified by pollution background, motivating a shift from spatial layout alone toward functionally adaptive design. Spatially matching green-space placement to pollution hotspots would enable differentiated strategies: around industrial areas, design “filtration” green belts using plant assemblages with low VOC emissions to form pollution-buffering zones, for example, Ginkgo biloba leaves have been shown to exhibit enhanced antioxidant capacity by increasing terpene synthesis when exposed to ozone pollution, thereby improving their stress resistance [31]. Other species have similar adaptive mechanisms through the production of secondary metabolites; Within residential neighborhoods, prioritize activity-oriented green spaces that promote physical activity and social interaction. This approach is consistent with the WHO's recent Healthy Cities/green-space guidance [32], and our empirical results provide specific species-selection and layout principles that can increase the guideline's operational utility.

Strengths and limitations

This study has several notable strengths, including its large sample size from two nationally representative cohorts ($n=12,903$), comprehensive exposure assessment using high-resolution environmental data, and sophisticated analytical approach examining interactions and multi-pollutant models. The inclusion of multiple sensitivity analyses further strengthens the robustness of our findings. However, several limitations should be acknowledged. The cross-sectional design limits causal inference, and exposure assessment relied on area-level averages rather than individual measurements. The composite cardiometabolic risk score, while validated, serves as a proxy for actual cardiovascular disease outcomes. Additionally, the spatial resolution of the environmental data, although derived from high-resolution datasets, still represents grid-based estimates which may not fully capture micro-environmental variations in individual exposure, potentially leading to exposure misclassification. Furthermore, NDVI alone cannot capture the quality, accessibility, or usage patterns of green spaces, and findings may have limited generalizability to regions with different environmental contexts. Despite these limitations, our study provides valuable insights into the complex interplay between environmental factors and cardiovascular health.

Conclusions

Based on an integrated analysis of the CHARLS and CLHLS databases, the results revealed a significant interaction between green space exposure and air pollutants in relation to cardiovascular disease risk. Overall, green space exposure showed a protective effect on cardiovascular health; however, this effect was strongly moderated by the level of air pollution. In low-pollution environments, higher exposure to green space effectively reduced cardiovascular risk, whereas in areas with severe pollution, the same exposure appeared to amplify the harmful effects of pollutants on the cardiovascular system—a finding that challenges conventional understanding.

The novelty of this research lies in uncovering the complex mechanisms underlying environmental health impacts and demonstrating the limitations of studying environmental factors in isolation. The interaction between NO₂ and green space exposure was particularly pronounced, suggesting that careful consideration is required when planning green space development in highly polluted regions. The results suggest that future environmental health research could benefit from examining green space and air pollution together rather than as separate domains.

These findings provide important implications for urban planning and public health policy. It is recommended that differentiated green space strategies be implemented in urban development, with tailored environmental interventions according to local pollution contexts. Future work should aim to establish dynamic environmental health risk assessment models that incorporate green space–pollution interactions into urban planning frameworks. These efforts will lay a scientific foundation for developing precise and tailored cardiovascular disease prevention strategies, helping to improve the world's health environment and address the increasingly severe burden of cardiovascular disease.

Abbreviations

NDVI	Normalized difference vegetation index
CHAP	China High-resolution Air Pollutant dataset
MetS	Metabolic syndrome
GLMs	Generalized linear models
IQR	Interquartile range
WHO	World Health Organization
CHARLS	China Health and Retirement Longitudinal Study
CLHLS	Chinese Longitudinal Healthy Longevity Survey
BMI	Body mass index
HDL-C	High-density lipoprotein cholesterol
GLMs	Generalized linear models
ORs	Odds ratios
IQR	Interquartile range
VOCs	Volatile organic compounds

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12889-026-27850-y>.

Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Supplementary Material 4.

Supplementary Material 5.

Supplementary Material 6.

Supplementary Material 7.

Supplementary Material 8.

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Authors' contributions

X.C. conceived this study and designed the research framework. Y.D. used CHARLS database and CLHLS database for statistical analysis and experiments. H. C. drafted the main manuscript and wrote the subject matter. Y.L. contributed to the development of methodology and the interpretation of data. S.Z. assisted data organization and verification. L.Z. assists in data collection and processing. M.Y. supervised the whole process of the project, guaranteed the funds, and made important amendments to the manuscript. All authors review and approve the final manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

In accordance with the Declaration of Helsinki, Peking University examined and authorized the studies involving human subjects in the China Health and Retirement Longitudinal Study (CHARLS) (IRB00001052–11015). The patients/ participants provided their written informed consent to participate in CHARLS. Our study used publicly available deidentified data from the CHARLS website (<http://charls.pku.edu.cn/index.htm>) and waived informed consent.

The Chinese Longitudinal Healthy Longevity Survey (CLHLS) provides free data to scholars, and the tracking survey data has been made available to society and academia for free after deleting personal privacy information. In addition to serving the scientific decision-making and application research of national health aging, it has also been accessed through the Peking University Open Research Data Platform (<http://opendata.pku.edu.cn/>) The "National Population Macro Management and Decision Information System (PADIS)" and "National Health Security Informatization Project" database platforms implemented by the China Population and Development Research Center are open for free use to academia and society.

Consent for publication

All authors have agreed to publish this paper.

Competing interests

The authors declare no competing interests.

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